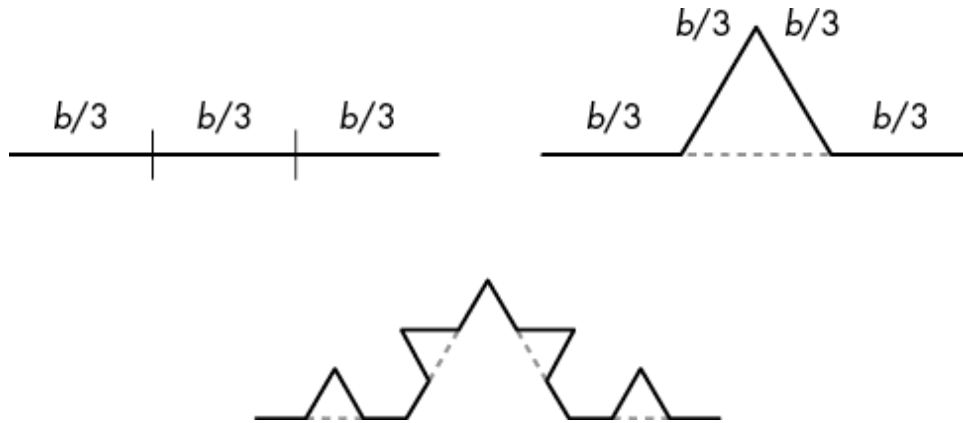
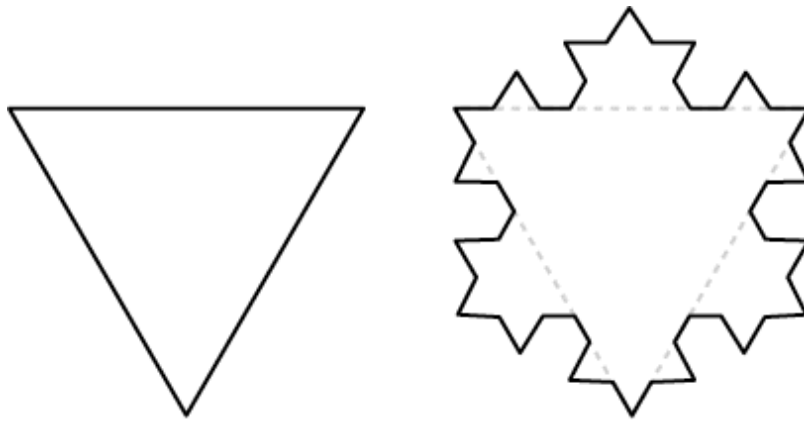


section with a “bump” whose sides are also of length $b / 3$. This bump causes the Koch curve to be longer than the original line, since we now have four line segments of length $b / 3$. (We’ll exclude the original middle part of the line segment.) This bump creation can be repeated on the new four line segments. [Figure 9-14](#) shows this construction.



[Figure 9-14](#): After splitting the line segment into three equal parts (left), add a bump to the middle part (right). We now have four segments of length $b / 3$, to which bumps can be added again (bottom).

To create the *Koch snowflake*, we start with an equilateral triangle and construct three Koch curves from its three sides, as in [Figure 9-15](#).



[Figure 9-15](#): Creating three Koch curves on the three sides of an equilateral triangle to form a Koch snowflake

Each time you create a new bump, you are increasing the curve’s length from three $b / 3$ lengths to four $b / 3$ lengths, or $4b / 3$. If you continue to do this with the three sides of an equilateral triangle, you’ll create the Koch snowflake, as in [Figure 9-16](#). (The small dotted patterns are artifacts,